

J4 Position vectors

Name: _____

Class: _____

Date: _____

Time: **474 minutes**

Marks: **392 marks**

Comments:

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q1.

At time $t = 0$, a parachutist jumps out of an airplane that is travelling horizontally. The velocity, $\mathbf{v}$ m s⁻¹, of the parachutist at time t seconds is given by:

$$\mathbf{v} = (40e^{-0.2t})\mathbf{i} + 50(e^{-0.2t} - 1)\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that the parachutist is at the origin when $t = 0$

Model the parachutist as a particle.

- (a) Find an expression for the position vector of the parachutist at time t .

(4)

- (b) The parachutist opens her parachute when she has travelled 100 metres horizontally.

Find the vertical displacement of the parachutist from the origin when she opens her parachute.

(4)

- (c) Carefully, explaining the steps that you take, deduce the value of g used in the formulation of this model.

(3)

(Total 11 marks)

Q2.

A helicopter travels at a constant height above the sea. It passes directly over a lighthouse with position vector $(500\mathbf{i} + 200\mathbf{j})$ metres relative to the origin, with a velocity of $(-17.5\mathbf{i} - 27\mathbf{j})$ m s⁻¹. The helicopter moves with a constant acceleration of $(0.5\mathbf{i} + 0.6\mathbf{j})$ m s⁻². The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the position vector of the helicopter t seconds after it has passed over the lighthouse.

(3)

- (b) The position vector of a rock is $(200\mathbf{i} - 400\mathbf{j})$ metres relative to the origin. Show that the helicopter passes directly over the rock, and state the time that it takes for the helicopter to move from the lighthouse to the rock.

(7)

- (c) Find the average velocity of the helicopter as it moves from the lighthouse to the rock.

(3)

- (d) Is the magnitude of the average velocity equal to the average speed of the helicopter?

Give a reason for your answer.

(2)

(Total 15 marks)

Q3.

A particle moves with a constant acceleration of $(0.1\mathbf{i} - 0.2\mathbf{j}) \text{ m s}^{-2}$. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the position vector of the particle t seconds after it has left the origin.

(2)

- (b) Find the time that it takes for the particle to reach the point where it is due east of the origin.

(3)

- (c) Find the speed of the particle when it is travelling south-east.

(6)

(Total 11 marks)

Q4.

A particle moves with a constant acceleration of $(0.1\mathbf{i} - 0.2\mathbf{j}) \text{ m s}^{-2}$. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the position vector of the particle t seconds after it has left the origin.

(2)

- (b) Find the time that it takes for the particle to reach the point where it is due east of the origin.

(3)

- (c) Find the speed of the particle when it is travelling south-east.

(6)

(Total 11 marks)

©2025 Exam Papers Practice. All rights reserved.

Q5.

A particle moves with constant acceleration $(-0.4\mathbf{i} + 0.2\mathbf{j}) \text{ m s}^{-2}$. Initially, it has velocity $(4\mathbf{i} + 0.5\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the velocity of the particle at time t seconds.

(2)

- (b) (i) Find the velocity of the particle when $t = 22.5$.

(2)

- (ii) State the direction in which the particle is travelling at this time.

(1)

- (c) Find the time when the speed of the particle is 5 m s^{-1} .

(6)

(Total 11 marks)

Q6.

The velocity of a particle at time t seconds is $\mathbf{v} \text{ m s}^{-1}$, where

$$\mathbf{v} = (4 + 3t^2)\mathbf{i} + (12 - 8t)\mathbf{j}$$

- (a) When $t = 0$, the particle is at the point with position vector $(5\mathbf{i} - 7\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(4)

- (b) Find the acceleration of the particle at time t .

(2)

- (c) The particle has mass 2 kg.

Find the magnitude of the force acting on the particle when $t = 1$.

(4)

(Total 10 marks)

Q7.

A helicopter is initially hovering above a lighthouse. It then sets off so that its acceleration is $(0.5\mathbf{i} + 0.375\mathbf{j}) \text{ m s}^{-2}$. The helicopter does not change its height above sea level as it moves. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the speed of the helicopter 20 seconds after it leaves its position above the lighthouse.

(4)

- (b) Find the bearing on which the helicopter is travelling, giving your answer to the nearest degree.

(3)

- (c) The helicopter stops accelerating when it is 500 metres from its initial position.

Find the time that it takes for the helicopter to travel from its initial position to the point where it stops accelerating.

(5)

(Total 12 marks)

Q8.

A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the velocity of the particle, $\mathbf{v} \text{ m s}^{-1}$, is given by

$$\mathbf{v} = 4e^{-2t}\mathbf{i} + (6t - 3t^2)\mathbf{j}$$

- (a) Find an expression for the acceleration of the particle at time t .

(3)

- (b) The mass of the particle is 5 kg.

- (i) Find an expression for the force $\mathbf{F}$ acting on the particle at time t . (2)
 - (ii) Find the magnitude of $\mathbf{F}$ when $t = 0$. (2)
 - (c) Find the value of t when $\mathbf{F}$ acts due west. (2)
 - (d) When $t = 0$, the particle is at the point with position vector $(6\mathbf{i} + 5\mathbf{j})$ m.
Find the position vector, $\mathbf{r}$ metres, of the particle at time t . (5)
- (Total 14 marks)

Q9.

Two forces, $\mathbf{P} = (6\mathbf{i} - 3\mathbf{j})$ newtons and $\mathbf{Q} = (3\mathbf{i} + 15\mathbf{j})$ newtons, act on a particle. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Find the resultant of $\mathbf{P}$ and $\mathbf{Q}$. (2)
- (b) Calculate the magnitude of the resultant of $\mathbf{P}$ and $\mathbf{Q}$. (2)
- (c) When these two forces act on the particle, it has an acceleration of $(1.5\mathbf{i} + 2\mathbf{j})$ m s⁻². Find the mass of the particle. (2)
- (d) The particle was initially at rest at the origin.
 - (i) Find an expression for the position vector of the particle when the forces have been applied to the particle for t seconds. (2)
 - (ii) Find the distance of the particle from the origin when the forces have been applied to the particle for 2 seconds. (2)

(Total 10 marks)

Q10.

A particle moves on a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

At time t seconds, the position vector of the particle is $\mathbf{r}$ metres, where

$$\mathbf{r} = \left(2e^{\frac{1}{2}t} - 8t + 5 \right) \mathbf{i} + (t^2 - 6t) \mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t . (3)
- (b) (i) Find the speed of the particle when $t = 3$. (2)
- (ii) State the direction in which the particle is travelling when $t = 3$. (1)
- (c) Find the acceleration of the particle when $t = 3$. (3)
- (d) The mass of the particle is 7 kg.
- Find the magnitude of the resultant force on the particle when $t = 3$. (3)
- (Total 12 marks)**

Q11.

A particle moves on a smooth horizontal plane. It is initially at the point A , with position vector $(9\mathbf{i} + 7\mathbf{j})$ m, and has velocity $(-2\mathbf{i} + 2\mathbf{j})$ m s⁻¹. The particle moves with a constant acceleration of $(0.25\mathbf{i} + 0.3\mathbf{j})$ m s⁻² for 20 seconds until it reaches the point B . The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the velocity of the particle at the point B . (3)
- (b) Find the velocity of the particle when it is travelling due north. (4)
- (c) Find the position vector of the point B . (3)
- (d) Find the average velocity of the particle as it moves from A to B . (2)
- (Total 12 marks)**

Q12.

A Jet Ski is at the origin and is travelling due north at 5 m s⁻¹ when it begins to accelerate uniformly. After accelerating for 40 seconds, it is travelling due east at 10 m s⁻¹. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the acceleration of the Jet Ski is $(0.1\mathbf{i} - 0.125\mathbf{j})$ m s⁻². (4)
- (b) Find the position vector of the Jet Ski at the end of the 40 second period. (3)

(c) The Jet Ski is travelling southeast t seconds after it leaves the origin.

(i) Find t .

(5)

(ii) Find the velocity of the Jet Ski at this time.

(2)

(Total 14 marks)

Q13.

A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the position vector, $\mathbf{r}$ metres, of the particle is given by

$$\mathbf{r} = (t^3 - 3t^2 + 4)\mathbf{i} + (4t + t^2)\mathbf{j}$$

(a) Find an expression for the velocity of the particle at time t .

(2)

(b) The mass of the particle is 3 kg.

(i) Find an expression for $\mathbf{F}$ at time t .

(3)

(ii) Find the magnitude of $\mathbf{F}$ when $t = 3$.

(2)

(c) Find the value of t when $\mathbf{F}$ acts due north.

(2)

(Total 9 marks)

Q14.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. A helicopter moves horizontally with a constant acceleration of $(-0.4\mathbf{i} + 0.5\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$, the helicopter is at the origin and has velocity $20\mathbf{i} \text{ m s}^{-1}$.

(a) Write down an expression for the velocity of the helicopter at time t seconds.

(2)

(b) Find the time when the helicopter is travelling due north.

(3)

(c) Find an expression for the position vector of the helicopter at time t seconds.

(2)

(d) When $t = 100$:

(i) show that the helicopter is due north of the origin;

(3)

- (ii) find the speed of the helicopter.

(3)

(Total 13 marks)

Q15.

A particle is initially at the origin, where it has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. It moves with a constant acceleration $\mathbf{a} \text{ m s}^{-2}$ for 10 seconds to the point with position vector $75\mathbf{i}$ metres.

- (a) Show that $\mathbf{a} = 0.5\mathbf{i} + 0.4\mathbf{j}$.

(3)

- (b) Find the position vector of the particle 8 seconds after it has left the origin.

(3)

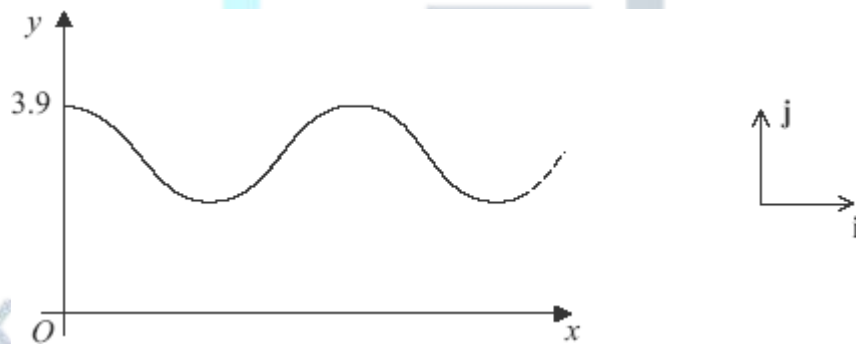
- (c) Find the position vector of the particle when it is travelling parallel to the unit vector $\mathbf{i}$.

(6)

(Total 12 marks)

Q16.

Jane is on a ride in a theme park. Part of the curved path of the ride is shown in the diagram.



Jane's position vector, $\mathbf{r}$ metres, at time t seconds is given by

$$\mathbf{r} = 1.2t \mathbf{i} + (3 + 0.9 \cos t) \mathbf{j}$$

where the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed horizontally and vertically upwards respectively.

- (a) Find an expression for Jane's velocity at time t .

(2)

- (b) (i) Find an expression for Jane's speed at time t .

(2)

- (ii) Find Jane's maximum speed during the ride.

(2)

(Total 6 marks)

Q17.

A particle has mass 800 kg. A single force of $(2400\mathbf{i} - 4800t\mathbf{j})$ newtons acts on the particle at time t seconds. No other forces act on the particle.

- (a) Find the acceleration of the particle at time t .

(2)

- (b) At time $t = 0$, the velocity of the particle is $(6\mathbf{i} + 30\mathbf{j}) \text{ m s}^{-1}$. The velocity of the particle at time t is $\mathbf{v} \text{ m s}^{-1}$.

Show that

$$\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 3t^2)\mathbf{j}$$

(4)

- (c) Initially, the particle is at the point with position vector $(2\mathbf{i} + 5\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(5)

(Total 11 marks)

Q18.

The points A and B have position vectors $(3\mathbf{i} + 2\mathbf{j})$ metres and $(6\mathbf{i} - 4\mathbf{j})$ metres respectively. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in a horizontal plane.

- (a) A particle moves from A to B with constant velocity $(\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. Calculate the time that the particle takes to move from A to B .

(3)

- (b) The particle then moves from B to a point C with a constant acceleration of $2\mathbf{j} \text{ m s}^{-2}$. It takes 4 seconds to move from B to C .

- (i) Find the position vector of C .

(4)

- (ii) Find the distance AC .

(2)

(Total 9 marks)

Q19.

A particle moves in a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = (2t^3 - t^2 + 6)\mathbf{i} + (8 - 4t^3 + t)\mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t .

(3)

- (b) (i) Find the velocity of the particle when $t = \frac{1}{3}$. (2)
- (ii) State the direction in which the particle is travelling at this time. (1)
- (c) Find the acceleration of the particle when $t = 4$. (3)
- (d) The mass of the particle is 6 kg. Find the magnitude of the resultant force on the particle when $t = 4$. (3)
- (Total 12 marks)**

Q20.

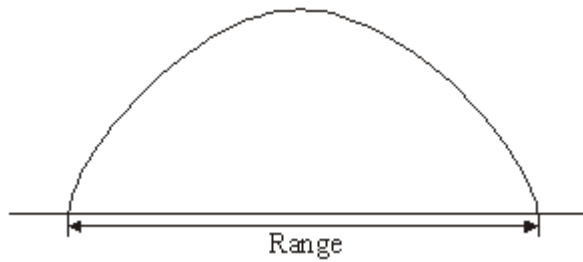
The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. A yacht moves with a constant acceleration. At time t seconds the position vector of the yacht is $\mathbf{r}$ metres. When $t = 0$ the velocity of the yacht is $(2\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$, and when $t = 10$ the velocity of the yacht is $(-\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$.

- (a) Find the acceleration of the yacht. (3)
- (b) When $t = 0$ the yacht is 20 metres due east of the origin. Find an expression for $\mathbf{r}$ in terms of t . (3)
- (c) (i) Show that when $t = 20$ the yacht is due north of the origin. (2)
- (ii) Find the speed of the yacht when $t = 20$. (4)
- (Total 12 marks)**

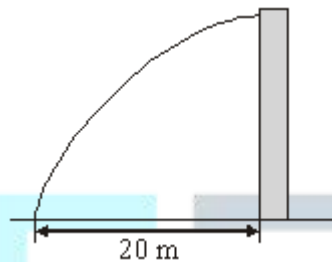
Q21.

A ball is hit from horizontal ground with velocity $(10\mathbf{i} + 24.5\mathbf{j}) \text{ m s}^{-1}$ where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertically upwards respectively.

- (a) State two assumptions that you should make about the ball in order to make predictions about its motion. (2)
- (b) The path of the ball is shown in the diagram.



- (i) Show that the time of flight of the ball is 5 seconds. (4)
- (ii) Find the range of the ball. (2)
- (c) In fact the ball hits a vertical wall that is 20 metres from the initial position of the ball.



- Find the height of the ball when it hits the wall. (4)
- (d) If a heavier ball were projected in the same way, would your answers to part (b) of this question change? Explain why. (2)

(Total 14 marks)

Q22.

A particle P , of mass 0.5 kg , moves under the action of a force $\mathbf{F}$ newtons, where $\mathbf{F} = 2t\mathbf{i} + 1.5\mathbf{j}$ at time t seconds.

- (a) Find the acceleration of P at time t . (1)
- (b) The particle P starts its motion from rest.
Find the velocity of P at time t . (2)
- (c) The particle P starts its motion at the point O .
- (i) Find the position vector of P relative to O at time t . (3)

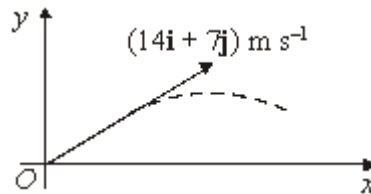
- (ii) Find the distance of P from O when it is moving parallel to the vector $\mathbf{i} + \mathbf{j}$.

(5)

(Total 11 marks)

Q23.

Jonny is practising kicking a rugby ball. He kicks the ball from a point O on horizontal ground with velocity $14\mathbf{i} + 7\mathbf{j} \text{ m s}^{-1}$. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in the directions of axes Ox and Oy , and Oy is vertically upwards.



- (a) (i) Find the coordinates of the ball t seconds after it has been kicked.

(3)

- (ii) Show that the equation of the path of the ball is

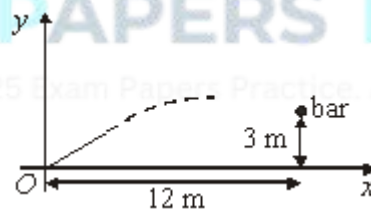
$$y = \frac{20x - x^2}{40}$$

(3)

- (iii) Find the horizontal range of the ball.

(2)

- (b) The ball moves towards a horizontal bar which is 3 metres above the ground and at a horizontal distance of 12 metres from O , as shown in the diagram.



Determine whether the ball passes above or below the bar.

(3)

- (c) State one modelling assumption that you have used in answering this question.

(1)

(Total 12 marks)

Q24.

A particle P moves so that at time t seconds its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 6t\mathbf{i} + t^2\mathbf{j}.$$

- (a) Find the velocity of P at time t . (2)
- (b) Find an expression for the speed of P at time t . (2)
- (c) Find the value of t when P moves with speed $6\sqrt{2}$ m s⁻¹. (2)
- (Total 6 marks)

Q25.

A particle P moves so that at time t it has velocity

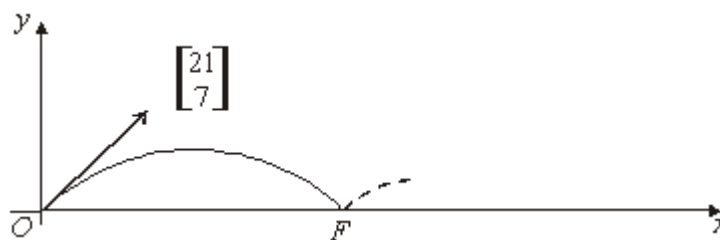
$$\mathbf{v} = 4\mathbf{i} - 2t\mathbf{j}, \quad t \geq 0.$$

When $t = 0$, the particle P has position vector $8\mathbf{j}$.

- (a) Find the position vector of P at time t . (4)
- (b) Find the position vector of P when $t = 2$. (1)
- (c) Find the time when the position vector of P is in the direction of the vector $\mathbf{i}$. (2)
- (Total 7 marks)

Q26.

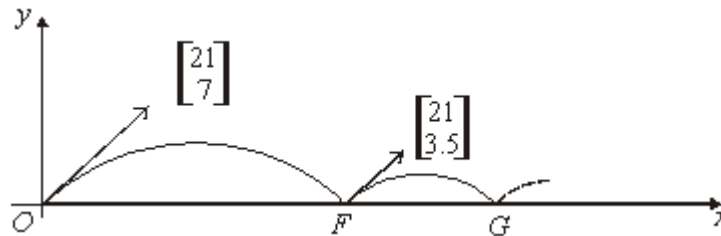
A ball is kicked from a point O on horizontal ground with velocity $\begin{bmatrix} 21 \\ 7 \end{bmatrix}$ m s⁻¹.
 The ball subsequently hits the ground at a point F , as shown in the diagram.



- (a) (i) Find the time between the ball being kicked and reaching F . (3)
- (ii) Find the distance OF . (2)

- (b) The ball rebounds from F with velocity $\begin{bmatrix} 21 \\ 3.5 \end{bmatrix}$ m s⁻¹. It subsequently hits the ground again at

G and then rebounds, as shown in the diagram below.



Find the distance FG .

(4)

- (c) After bouncing at G , the ball rebounds with velocity $\begin{bmatrix} 21 \\ 1.75 \end{bmatrix}$ m s⁻¹, so that the vertical component of the velocity is halved again, and then subsequently bounces next at H .

Find the total distance OH .

(3)

(Total 12 marks)

Q27.

A particle P moves so that at time t seconds its velocity $\mathbf{v}$ m s⁻¹ is

$$\mathbf{v} = 2t\mathbf{i} + 4\mathbf{j}, \quad t \geq 0.$$

- (a) At time $t = 0$, the particle is at the point with position vector $2\mathbf{j}$ metres.

Find the position vector of P at time t .

(4)

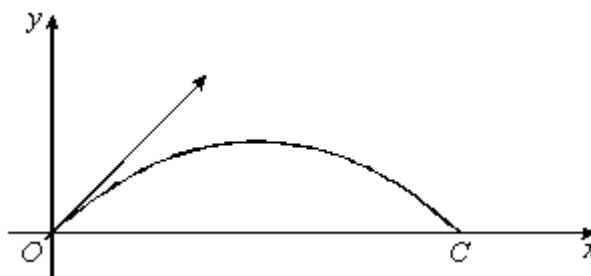
- (b) At times $t = 2$ and $t = 4$, the particle passes through the points C and D respectively.

Find the vector $\overrightarrow{CD}$.

(4)

(Total 8 marks)

Q28.



Oliver throws a ball from point O , with velocity $\begin{bmatrix} 7 \\ 7 \end{bmatrix}$ m s⁻¹. The ball is subsequently caught by

Chris at a point C which is on the same horizontal level as O , as shown in the diagram. During the motion from O to C the ball moves in a vertical plane.

- (a) Taking O as the origin, find the position vector of the ball t seconds after being thrown.

(3)

- (b) Show that the equation of the path of the ball is

$$y = x - 0.1x^2,$$

where (x, y) are the coordinates at time t of the ball relative to the origin O .

(4)

- (c) A horizontal wire, which is part of a fence, cuts the plane of motion of the ball at the point $(4, 2)$.

Determine whether the ball passes above or below the wire.

(3)

- (d) Find the distance OC .

(4)

- (e) Find the greatest speed of the ball during its motion from O to C .

(2)

(Total 16 marks)

Q29.

At time t , the position vector of a particle Q is

$$\mathbf{r} = (t^2 - 6t + 4)\mathbf{i} + \frac{1}{3}t^3\mathbf{j}.$$

- (a) Find the velocity of Q at time t .

(3)

- (b) Find the value of t when Q is moving parallel to the vector $\mathbf{j}$.

(2)

- (c) Find the acceleration of Q and state, with a reason, whether or not this acceleration is constant.

(2)

(Total 7 marks)

Q30.

A particle moves with constant acceleration in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the plane. At time $t = 0$, the particle is at rest at the origin. At time $t = 4$ seconds, the position vector of the particle is $(-16\mathbf{i} + 16\mathbf{j})$ metres.

- (a) Show that the acceleration of the particle is $(-2\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$.
- (b) Find the speed of the particle when $t = 5$.
- (c) The mass of the particle is 3 kg. Two horizontal forces $\mathbf{P}$ and $\mathbf{Q}$ act on the particle. If $\mathbf{P} = 20\mathbf{i} - 10\mathbf{j}$, find $\mathbf{Q}$.

(4)

(4)

(4)

(Total 12 marks)

Q31.

A particle moves with constant acceleration $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 10\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the velocity of the particle at time t seconds is

$$(3 + 4t)\mathbf{i} + (2t - 10)\mathbf{j}$$

(2)

- (b) Find the time when the particle is travelling due east.

(2)

- (c) (i) Find the position vector of the particle when $t = 10$.

(3)

- (ii) Find the velocity of the particle when $t = 10$.

(1)

- (d) The particle stops accelerating when $t = 10$ and then moves with constant velocity.

Find the distance of the particle from the origin after it has been travelling with this constant velocity for a further 10 seconds.

(6)

(Total 14 marks)

Q32.

A particle is initially at the origin and has velocity $(4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. It moves with constant acceleration, and 10 seconds later has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the acceleration of the particle is $(0.1\mathbf{i} - 0.5\mathbf{j}) \text{ m s}^{-2}$. (3)
 - (b) Find an expression for the position vector of the particle at time t seconds. (2)
 - (c) Find the time when the particle is due east of the origin. (4)
 - (d) Find the time when the particle is moving due east. (4)
- (Total 13 marks)

Q33.

At time $t = 0$, a boat has velocity $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$ and is at the origin. After moving with a constant acceleration of $\mathbf{a} \text{ m s}^{-2}$ for 20 seconds, its position vector is $(60\mathbf{i} + 20\mathbf{j})$ metres. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find $\mathbf{a}$. (3)
 - (b) Find the velocity of the boat at time t seconds. (2)
 - (c) Find the speed of the boat when it is travelling south east. (5)
- (Total 10 marks)

Q34.

A particle moves in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Initially the particle is at the origin and has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. It moves with constant acceleration. At time $t = 10$ it has position vector $(40\mathbf{i} - 15\mathbf{j})$ metres.

- (a) Show that the velocity of the particle is $(3\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$ when $t = 10$. (4)
- (b) Find the acceleration of the particle. (3)
- (c) The mass of the particle is 15 kg. Find the magnitude of the resultant force acting on the particle. (4)

(Total 11 marks)

Q35.

A particle of mass 4 kg moves on a smooth horizontal plane. It is initially at rest at the origin. A force $\mathbf{F} = (8\mathbf{i} - 12\mathbf{j})$ N acts on the particle for 20 seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

- (a) (i) Find the acceleration of the particle. (2)
- (ii) Find the velocity of the particle at the end of the 20 second period. (2)
- (iii) Find the position of the particle at the end of the 20 second period. (3)
- (b) At the end of the 20 second period, the force $\mathbf{F}$ is removed. Find the distance of the particle from the origin after the particle has been in motion for a total of 45 seconds. (5)

(Total 12 marks)